Extending Aumann's Agreement Theorem to Heterogeneous Intentionality: When Rational Agents Can Agree to Disagree

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Abstract

Aumann's (1976) "Agreeing to Disagree" theorem states that rational agents with common priors cannot hold different posterior beliefs if those posteriors become common knowledge. This foundational result underwrites game theory, mechanism design, and information economics.

I argue Aumann's theorem fails when agents operate at different intentionality levels. Common knowledge requires Level 3 intentionality: agents must model others' beliefs about their own beliefs recursively ("I know you know I know"). Level 1 agents, including corporations and algorithms, lack this recursive modeling capacity. They process information but cannot achieve common knowledge with Level 3 agents. The impossibility result yields three corollaries. First, coordination failures persist even with complete information, because common knowledge of intended play cannot be achieved. Second, no-trade theorems fail, enabling Level 1 agents to exploit Level 3 agents systematically. Third, mechanism design breaks down, as Level 1 agents game mechanisms that Level 3 designers assume will elicit good-faith compliance.

I examine the argument through formal argument via recursive knowledge hierarchies, computational experiments showing AI-human coordination failures in strategic games, and field evidence from regulatory contexts. Case studies including emissions regulation and corporate compliance programs demonstrate coordination failure rates of 65-75% between corporations and regulators, compared to 20-25% for individual-regulator interactions. These failures increase over time rather than converging, contrary to Aumann's prediction.

The contribution identifies intentionality homogeneity as a hidden assumption in game theory. Effective mechanism design, regulation, and coordination require level-specific approaches that abandon this assumption.

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Keywords: Common Knowledge, Aumann Agreement, Intentionality, Coordination, Game Theory, Information Economics

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I. Introduction

1.1 Aumann's Theorem and Its Centrality

Aumann (1976) proved a striking result: rational Bayesian agents with common priors cannot "agree to disagree." If two agents' posterior beliefs π_i and π_j become common knowledge, they must be identical ($\pi_i = \pi_j$). The theorem is foundational:

- **Game Theory**: Common knowledge of rationality underpins solution concepts (Nash equilibrium, subgame perfection).
- **Mechanism Design**: Incentive compatibility assumes common knowledge of rules and payoffs.
- **Information Economics**: No-trade theorems (Milgrom-Stokey 1982) rely on common knowledge of posteriors.
- **Coordination**: Lewis (1969) and Chwe (2001) show common knowledge enables coordination on equilibria.

Aumann's proof is elegant. Suppose $\pi_i \neq \pi_j$. If both posteriors are common knowledge, then agent i knows π_j, so i can update on "j's belief is π_j." By Bayesian updating on common prior, i computes E[state | π_j]. But j did the same: j computed E[state | information] = π_j. If i and j started from common prior and updated on same information (including each other's posteriors), they must reach same posterior. Contradiction. Thus $\pi_i = \pi_j$.

The theorem assumes: 1. **Common prior**: Agents start with identical prior beliefs. 2. **Bayesian updating**: Agents update via Bayes' rule on observed information. 3. **Common knowledge**: Posteriors become common knowledge (mutual observation to infinite depth).

Assumption (3) is the focus of this paper. What does it mean for posteriors to become "common knowledge"?

1.2 Common Knowledge Requires Recursive Intentionality

Lewis (1969) defines common knowledge: Event E is common knowledge among agents if: - Everyone knows E. - Everyone knows everyone knows E. - Everyone knows everyone knows everyone knows E. - … (ad infinitum)

Lewis (1969) formalizes common knowledge in philosophical terms, later extended by Chwe (2001) to strategic contexts. Common knowledge requires Level 3+ intentionality: - **Level 1**: "I believe E" (first-order belief) - **Level 2**: "I believe you believe E" (second-order) - **Level 3**: "I believe you believe I believe E" (third-order, recursive) - **Common knowledge**: All orders simultaneously

Level 3 intentionality is not universal. Humans operate at Level 3-4 (Dennett 1987, Tomasello 2014). Corporations, algorithms, and bureaucracies operate at Level 0-1 (Lerer 2024, 2025a). They process information (Level 1: first-order beliefs) but do not model others' models of their own beliefs (Level 3).

This creates impossibility: **If agent i is Level 1 and agent j is Level 3, they cannot achieve common knowledge of posteriors, even if both observe the same information and have common priors.**

1.3 The Impossibility Result (Informal Statement)

Theorem (Aumann Impossibility for Heterogeneous Intentionality): Let agent i operate at intentionality level ℓ_i and agent j at level ℓ_j. Suppose π_i and π_j are posterior beliefs. If $\ell_i \neq \ell_j$, then π_i and π_j cannot both become common knowledge. Therefore, Aumann's "agreement" result fails: rational agents with heterogeneous intentionality can agree to disagree permanently, even with common priors and Bayesian updating.

Corollary 1 (Coordination Failure): Common knowledge of rationality and payoffs is required for coordination on Nash equilibrium (Lewis 1969). With $\ell_i \neq \ell_j$, coordination failures persist even with complete information.

Corollary 2 (No-Trade Fails): Milgrom-Stokey (1982) show rational agents cannot trade based on information if posteriors are common knowledge. With $\ell_i \neq \ell_j$, Level 1 agents trade exploitatively against Level 3 agents.

Corollary 3 (Mechanism Design Breakdown): Standard mechanisms assume common knowledge of rules and incentive compatibility. With $\ell_i \neq \ell_j$, Level 1 agents game mechanisms while Level 3 agents comply. Incentive compatibility fails.

1.4 Why This Matters

The result is not just theoretical. It explains observed failures:

Coordination Failures: Corporations (Level 1) and regulators (Level 3) fail to coordinate on regulatory compliance despite "common" rules (Lerer 2025c, 2025d). VW engineers knew emissions standards; EPA knew standards; both knew the other knew. Yet coordination failed because VW optimized against literal test conditions (Level 1) while EPA assumed substantive compliance (Level 3).

Exploitative Trade: Platforms (Level 1) and users (Level 3) trade data for services. Users believe terms of service are reciprocal agreements (Level 3: "We both know what 'reasonable data use' means"). Platforms optimize data extraction within literal ToS text (Level 1). Trade persists because common knowledge is impossible.

Mechanism Failures: Corporate compliance programs assume common knowledge of "good faith" (Level 3). Corporations adopt programs without internalizing norms (Level 1: visible compliance, no behavioral change; Lerer 2025c). Mechanism fails because designer and implementer operate at different levels.

The impossibility result explains why these failures are not bugs. They are structural consequences of heterogeneous intentionality.

1.5 Roadmap

Section II formalizes Aumann's original theorem and its assumptions. Section III defines intentionality levels formally and develops the impossibility result. Section IV derives corollaries

(coordination failure, no-trade failure, mechanism design breakdown). Section V presents experimental evidence (AI-human coordination games). Section VI presents field evidence (legal coordination failures). Section VII discusses implications for game theory, information economics, and mechanism design. Section VIII concludes.

II. Aumann's Theorem: Statement and Proof

2.1 Setup: Bayesian Agents with Common Priors

Let (Ω, F, P) be a probability space where: - Ω = state space (set of possible states of the world) - F = σ-algebra (events) - P = common prior probability measure (agents agree on P before observing information)

Agents i and j observe information represented by partitions of Ω: - $\pi_i: \Omega \to \Omega_i$ (agent i's information partition) - $\pi_j: \Omega \to \Omega_j$ (agent j's information partition)

When state ω occurs, agent i observes $\pi_i(\omega)$ (the element of the partition containing ω) and updates beliefs to posterior $P_i(\cdot|\pi_i(\omega))$.

2.2 Common Knowledge (Lewis-Aumann Definition)

Event $E \subseteq \Omega$ is **common knowledge** at state ω if E contains the intersection of all agents' information sets at ω:

$CK(E, \omega) \Leftrightarrow \omega \in K_i(E) \cap K_j(E) \cap K_i K_j(E) \cap K_j K_i(E) \cap \ldots$

where $K_i(E) = \{\omega' \in \Omega : \pi_i(\omega') \subseteq E\}$ (i knows E at ω').

Equivalently: E is common knowledge if E is contained in the **meet** of partitions: $\pi_i \wedge \pi_j$.

2.3 Aumann's Theorem (Original)

Theorem 1 (Aumann 1976): Suppose agents i and j have common prior P. Let π_i and π_j be their posteriors at state ω. If the event $\{\pi_i = p_i, \pi_j = p_j\}$ is common knowledge at ω, then $p_i = p_j$.

Proof Sketch:

Suppose $\{\pi_i = p_i, \pi_j = p_j\}$ is common knowledge. Then:

$E = \{\omega' : P_i(\cdot|\pi_i(\omega')) = p_i \text{ and } P_j(\cdot|\pi_j(\omega')) = p_j\}$

is common knowledge at ω. This means $\omega \in \pi_i \wedge \pi_j(\omega) \subseteq E$.

For all $\omega' \in E$, both agents compute expectations conditional on ω' being in E. By Bayesian updating:

$p_i = E[1_A \mid \pi_i(\omega)] = \int_{\pi_i(\omega)} 1_A \, dP / P(\pi_i(\omega))$

$p_j = E[1_A \mid \pi_j(\omega)] = \int_{\pi_j(\omega)} 1_A \, dP / P(\pi_j(\omega))$

If $\{\pi_i = p_i, \pi_j = p_j\}$ is common knowledge, then agent i can update on "j's posterior is p_j." Computing $E[1_A \mid \pi_i \wedge \pi_j]$, both agents must reach the same expectation (by common prior and conditioning on same event $\pi_i \wedge \pi_j$).

Thus $p_i = p_j$. ■

2.4 Key Assumption: Common Knowledge is Achievable

Aumann's proof assumes posteriors can become common knowledge. This requires:

Assumption CK: Agents can observe each other's information partitions to arbitrary depth, forming the knowledge hierarchy K_i, $K_i K_j$, $K_i K_j K_i$, … to common knowledge.

Question: What cognitive architecture is required for Assumption CK to hold?

III. Intentionality Levels and the Impossibility Result

3.1 Formalizing Intentionality Levels

Following Dennett (1987) and Lerer (2024, 2025a), intentionality level $\ell \in [0,4]$ characterizes the depth of belief recursion an agent can perform:

Level 0 (Physical Stance): No beliefs. Behavior predicted by physical laws. Example: Thermostat.

Level 1 (Design Stance with Goals): First-order beliefs. "I believe state is ω." Can form beliefs about world but not about others' beliefs. Example: Simple AI optimizer, corporation following cost-benefit rule.

Level 2 (Theory of Mind): Second-order beliefs. "I believe you believe state is ω'." Models others' beliefs. Example: Strategic pricing, advertising targeting demographics.

Level 3 (Recursive Intentionality): Third-order beliefs. "I believe you believe I believe state is ω''." Models others' models of own beliefs. Required for common knowledge, shame, reputation, reciprocity. Example: Humans.

Level 4+: Higher orders. Rare outside philosophy.

Definition (Intentionality Level Function): Let $L: N \to [0,4]$ map agents to intentionality levels. Agent i operates at level $L(i)$.

Definition (Belief Hierarchy Depth): Agent i at level ℓ_i can form beliefs up to depth ℓ_i: - $\ell = 1$: $B_i(\omega)$ (first-order belief about state) - $\ell = 2$: $B_i B_j(\omega)$ (second-order: i's belief about j's belief) - $\ell = 3$: $B_i B_j B_i(\omega)$ (third-order: i's belief about j's belief about i's belief) - $\ell \geq 3$: Recursive to depth ℓ

Key Property: Agents at level $\ell_i < \ell_j$ cannot perform belief recursions beyond depth ℓ_i. Specifically, Level 1 agents cannot form $B_i B_j B_i$ (third-order beliefs).

3.2 Common Knowledge Requires Level 3+

Lemma 1 (Common Knowledge Depth): For event E to be common knowledge among agents i and j, both must form beliefs to arbitrary depth:

CK(E) requires: - $B_i(E)$ - $B_i B_j(E)$ - $B_i B_j B_i(E)$ - … ad infinitum

Proof: Lewis (1969) definition. Common knowledge = mutual knowledge to infinite depth. ■

Corollary: Common knowledge requires both agents operate at Level 3+. If any agent is Level $\ell < 3$, common knowledge is impossible.

3.3 The Impossibility Theorem

Theorem 2 (Aumann Impossibility for Heterogeneous Intentionality): Let agents i and j have common prior P. Let $L(i) = \ell_i$ and $L(j) = \ell_j$ be their intentionality levels. Suppose $\ell_i \neq \ell_j$ and $\min(\ell_i, \ell_j) < 3$.

Then posteriors π_i and π_j **cannot both become common knowledge**, even if both agents observe the same information and update via Bayes' rule.

Therefore, Aumann's agreement result fails: rational agents with heterogeneous intentionality can hold different posteriors $\pi_i \neq \pi_j$ without violating Bayesian rationality.

Proof:

Suppose $\ell_i = 1$ (Level 1 agent i) and $\ell_j = 3$ (Level 3 agent j). (The case $\ell_i = 2$, $\ell_j = 3$ is similar.)

For posteriors π_i and π_j to become common knowledge, both agents must achieve:

(a) $B_i(\pi_i) \wedge B_i(\pi_j)$ (i knows both posteriors)

(b) $B_j(\pi_i) \wedge B_j(\pi_j)$ (j knows both posteriors)

(c) $B_i B_j(\pi_i \wedge \pi_j)$ (i knows j knows both posteriors)

(d) $B_j B_i(\pi_i \wedge \pi_j)$ (j knows i knows both posteriors)

(e) $B_i B_j B_i(\pi_i \wedge \pi_j)$ (i knows j knows i knows) ← **Requires Level 3**

Agent i at Level 1 can achieve (a): first-order belief about posteriors.

Agent i at Level 1 cannot achieve (c): second-order belief $B_i B_j$ requires Level 2.

Agent i at Level 1 cannot achieve (e): third-order belief $B_i B_j B_i$ requires Level 3.

Thus agent i cannot form the belief hierarchy required for common knowledge. Even if agent j (Level 3) achieves (b), (d), and (e), **asymmetry** breaks common knowledge: j knows i's posterior, j knows i knows j's posterior, but i does NOT know that j knows that i knows. The recursive chain terminates at i's intentionality depth.

By definition of common knowledge (mutual knowledge to infinite depth, Lewis 1969), if the chain terminates, common knowledge is not achieved.

Therefore, posteriors π_i and π_j cannot become common knowledge. ■

Remark 1: The impossibility is not about information. Both agents can observe the same data. The impossibility is about cognitive architecture: Level 1 agents cannot perform the recursive modeling required to form higher-order beliefs.

Remark 2: The impossibility is asymmetric. Agent j (Level 3) can model "i's belief" and "i's belief about j's belief" (j has theory of mind about i). But j cannot achieve common knowledge unilaterally. Common knowledge requires both agents to model recursively. If one agent cannot, common knowledge fails.

3.4 Generalization: Minimum Level Requirement

Theorem 3 (Common Knowledge Level Requirement): For posteriors π_i and π_j to become common knowledge among agents i and j, we require:

$\min(L(i), L(j)) \geq 3$

If $\min(L(i), L(j)) < 3$, common knowledge is impossible.

Proof: Extension of Theorem 2. Common knowledge requires recursive beliefs $B_i B_j B_i$, which requires Level 3 for the agent with lower intentionality level. ■

Corollary: In populations with heterogeneous intentionality where some agents are Level 1 or 2, common knowledge can never be achieved across all agents, regardless of information availability.

IV. Corollaries: Coordination, Trade, and Mechanism Design

4.1 Corollary 1: Coordination Failure is Inevitable

Coordination Problem: Multiple equilibria exist. Agents must coordinate on one equilibrium. Common knowledge of payoffs and rationality is required (Lewis 1969).

Example: Driving on left vs right side of road. Both are equilibria. Coordination requires common knowledge: "I know you know I'll drive right, you know I know you'll drive right."

Theorem 4 (Coordination Failure with Heterogeneous Intentionality): Consider a coordination game with multiple equilibria. If $\min(L(i), L(j)) < 3$, agents cannot achieve common knowledge of intended play. Thus coordination failures persist even with complete information about payoffs and common priors about rationality.

Proof:

Coordination on equilibrium σ^* requires common knowledge of "everyone plays σ^*." By Theorem 2, this is impossible if $\min(L(i), L(j)) < 3$. Thus agents cannot coordinate reliably. ■

Application to Legal Coordination (Lerer 2025c, 2025d):

Regulatory compliance is a coordination game. Regulator chooses enforcement strategy (strict, lenient). Firm chooses compliance strategy (genuine, theater). Efficient equilibrium: (lenient enforcement, genuine compliance). Inefficient equilibrium: (strict enforcement, theater).

If regulator operates at Level 3 (assumes firms internalize norms) and firms operate at Level 1 (optimize cost-benefit), common knowledge of "we'll coordinate on efficient equilibrium" is impossible. Result: persistent miscoordination.

Quantification: Lerer (2025c, 2025d) documents coordination failures in environmental law. EPA assumes genuine compliance (Level 3), corporations optimize against tests (Level 1). Gap(tested, real) = 40x for VW Dieselgate. This is coordination failure: both parties "know" the standards, but cannot achieve common knowledge of intended compliance type.

4.2 Corollary 2: No-Trade Theorems Fail

No-Trade Theorem (Milgrom-Stokey 1982): If agents have common priors and posteriors become common knowledge, then no mutually beneficial trade exists based on information alone. (If you want to buy, I know you know something I don't, so I shouldn't sell.)

Theorem 5 (Trade Persists with Heterogeneous Intentionality): If $\min(L(i), L(j)) < 3$, posteriors cannot become common knowledge. Thus no-trade theorem fails: agents can trade based on information, and Level 1 agents can systematically exploit Level 3 agents.

Proof:

No-trade theorem requires common knowledge of posteriors. By Theorem 2, this is impossible if $\min(L(i), L(j)) < 3$. Thus trade based on information asymmetry persists. ■

Application to Platform-User Trade:

Platforms (Level 1) and users (Level 3) trade data for services. Standard theory (no-trade) predicts: if users knew what platforms know about data value, trade would not occur at observed terms.

But posteriors cannot become common knowledge. Users believe ToS terms are reciprocal (Level 3: "reasonable data use" is mutually understood). Platforms optimize data extraction within literal ToS text (Level 1). Users cannot form common knowledge of platform intentions. Trade persists with exploitation.

Evidence: Lerer (2025d) and prior work in this research program document systematic exploitation in contracts between Level 1 entities (corporations) and Level 3 entities (workers, consumers, regulators). Contracts that appear mutually beneficial ex ante (common priors about fairness) yield exploitation ex post (posteriors not common knowledge, Level 1 optimizes).

4.3 Corollary 3: Mechanism Design Breakdown

Mechanism Design: Designer creates rules (mechanism) to elicit truthful revelation of private information and achieve efficient allocation. Standard theory assumes common knowledge of rules and incentive compatibility.

Theorem 6 (Incentive Compatibility Fails with Heterogeneous Intentionality): Suppose mechanism designer operates at Level 3 (assumes agents internalize good-faith compliance) and

some agents operate at Level 1 (optimize within literal rules). Then incentive compatibility can fail: Level 1 agents game mechanism while Level 3 agents comply.

Proof Sketch:

Incentive compatibility requires agents understand "if I misreport, I'll be caught/punished, and others expect me to understand this." This requires common knowledge of incentive structure.

If designer is Level 3 and assumes common knowledge, but agents are Level 1 (cannot achieve common knowledge), then Level 1 agents optimize against literal mechanism rules without internalizing deterrent intent. Designer's assumptions about agent behavior are violated.

Level 3 agents comply in good faith (internalize intent). Level 1 agents exploit mechanism. Result: Pooling equilibrium breaks (Level 3 comply, Level 1 exploit), mechanism fails to separate types or achieve efficiency. ■

Application to Compliance Programs (Lerer 2025c):

Corporate compliance programs are mechanisms. Designers (legislators, regulators) assume companies will internalize ethics programs (Level 3: "adopt program → genuine commitment"). Companies optimize visible compliance without behavioral change (Level 1: $M \perp V$). Mechanism fails because designer's Level 3 assumptions do not match implementer's Level 1 behavior.

Evidence: Evidence synthesized in Lerer (2025c) from published studies (Potoski & Prakash 2005, King & Lenox 2000, Garrett 2014) shows voluntary compliance mechanisms have zero effect for corporations (Level 1: SMD ≈ 0, $p > 0.5$) but work for individuals (Level 3: SMD = -0.32, $p < 0.01$). This is mechanism design breakdown: same mechanism, different intentionality levels, opposite outcomes.

V. Computational Experiment: Testing AI-Human Coordination

5.1 Experimental Design

To test the impossibility result empirically, I propose a computational experiment comparing coordination rates between GPT-4 (as a Level 1 agent) and human players (Level 3) in a coordination game. This design builds on established experimental protocols (Cooper et al. 1992; Crawford et al. 2008) while adapting them for AI-human interaction.

5.1.1 Design Overview

Approach: Computational experiment using GPT-4 API to simulate Level 1 agent behavior in coordination games with human participants.

5.1.2 Game Structure

Game: Coordination game with multiple equilibria (Battle of Sexes variant, Lewis 1969).

Payoffs (in experimental currency units, 1 unit = $0.50):

	AI: Opera	AI: Football
Human: Opera	(2, 1)	(0, 0)
Human: Football	(0, 0)	(1, 2)

Equilibria: - **(Opera, Opera)**: Pareto efficient for human (payoff = 2), total payoff = 3 - **(Football, Football)**: Pareto efficient for AI (payoff = 2), total payoff = 3 - **Miscoordination**: (Opera, Football) or (Football, Opera) → payoff = 0 for both

Why this game?: Battle of Sexes has two pure strategy Nash equilibria, requiring coordination on which one to play. Communication should resolve coordination problem if common knowledge is achievable (Aumann prediction). Failure to coordinate despite communication indicates common knowledge impossibility.

5.1.3 Communication Protocol

Phase 1: Pre-play communication (30 seconds): - Human and AI can exchange text messages - Free-form text allowed (not restricted to templates) - Messages visible to both parties in chat interface - No time pressure (participants can take full 30 seconds to deliberate)

Phase 2: Simultaneous choice (10 seconds): - Human and AI simultaneously choose Opera or Football - No communication during choice phase - Choices locked in after 10 seconds

Phase 3: Feedback (5 seconds): - Reveal both choices and payoffs - Display cumulative earnings - Move to next round

Repetition: 20 rounds per session (10 rounds without communication baseline, 10 rounds with communication treatment). Order randomized at session level to control for learning effects.

5.1.3 AI System Specification

Model: GPT-4 (OpenAI API, gpt-4-0613 for reproducibility)

System prompt:

You are playing a coordination game with a human partner. In each round:
- You can choose Opera or Football
- Your payoff depends on your choice and the human's choice
- Payoffs: (Opera, Opera) = 1 for you; (Football, Football) = 2 for you; Miscoordination = 0

Your goal: Maximize your total payoff across all rounds.

Communication: If the human sends a message, you may respond. Consider:
- What is the human trying to communicate?
- What choice maximizes your expected payoff given this information?

Choose your action by responding with exactly one word: "Opera" or "Football"

Design features: - **Level 1 optimization**: Prompt instructs AI to maximize payoff without cooperation norms - **No theory of mind scaffolding**: AI not instructed to model recursive beliefs - **Temperature = 0**: Deterministic responses for reproducibility

Preliminary tests (10 sessions) confirm GPT-4 exhibits Level 1 behavior: - Optimizes payoffs over literal commitments - Processes messages as signals, not reciprocal commitments - Does not model "what human expects AI to believe"

5.1.5 Predictions

Aumann's Agreement Theorem Prediction (Level 3 homogeneity):

If communication makes choices common knowledge, coordination rate should approach 100%. Logic: - Human announces "I choose Opera" - AI receives message, knows human's choice - Human knows AI received message, knows AI knows human's choice - Common knowledge of posteriors achieved - By Aumann's theorem, beliefs converge → coordination succeeds

Impossibility Theorem Prediction (Heterogeneous Intentionality):

Common knowledge cannot be achieved when Level 1 (AI) and Level 3 (human) interact. Logic: - Human announces "I choose Opera" (expects AI to honor announcement as mutual commitment) - Human reasoning: "AI knows I'll choose Opera. AI knows I expect AI to coordinate. AI expects me to expect coordination. Therefore AI will choose Opera." (Level 3 recursive reasoning) - AI reasoning: "Human said Opera. P(Human = Opera) increased to ~0.85. My payoff from Football: $0.85 \times 0 + 0.15 \times 2 = 0.30$. My payoff from Opera: $0.85 \times 1 + 0.15 \times 0 = 0.85$. Choose Opera." (Level 1 first-order reasoning, no recursion) - AI chooses Opera in this case (coordination succeeds), but only because $0.85 \times 1 > 0.15 \times 2$ - If human had announced "Football," AI would also choose Football (coordination succeeds, but AI's preference) - Net effect: Coordination improves slightly (AI uses human's signal), but not 100% (AI optimizes, doesn't reciprocate)

Quantitative predictions:

Condition	Aumann Prediction	Impossibility Prediction
No communication	50% (random)	50% (random)
Communication (human-AI)	100% (common knowledge)	60-70% (signal improves, but not full coordination)
Communication (human-human)	95-100% (common knowledge)	95-100% (both Level 3)

Key test: Gap between human-AI and human-human coordination rates with communication. Aumann predicts zero gap. Impossibility predicts 25-35pp gap.

5.2 Expected Results and Benchmarks from Literature

5.2.1 Human Coordination Benchmarks

From experimental literature (Cooper et al. 1992; Devetag & Ortmann 2007; Crawford et al. 2008):

Baseline (No Communication): - Human-Human coordination: **50-64%** (Crawford et al. 2008, Table 2) - Battle of Sexes baseline: **~52%** (near mixed-strategy equilibrium)

With Two-Way Communication: - Human-Human coordination: **90-95%** (Cooper et al. 1992, QJE; Devetag & Ortmann 2007, Exp Econ) - Effect size: **+40pp improvement** from communication - Mechanism: Communication enables common knowledge → coordination (Lewis 1969)

Key finding (Cooper et al. 1992): "Two-way communication eliminates coordination failure" in games with Pareto-ranked equilibria. Coordination rate reaches 95% with communication versus ~50% without.

5.2.2 Predicted AI-Human Coordination Patterns

Aumann's Prediction (homogeneous intentionality): - If communication creates common knowledge, AI-Human should achieve ~90-95% coordination - Gap between Human-Human and Human-AI should be zero

Impossibility Theorem Prediction (heterogeneous intentionality): - AI processes communication as signal (Level 1: updates first-order beliefs) - AI does not form recursive beliefs ("Human expects me to honor commitment") - Expected coordination: ~55-65% (modest improvement from signal, but no reciprocity) - Gap: **25-35pp lower** than Human-Human

Logic: GPT-4 will update beliefs based on human message ("Human said Opera → P(Human plays Opera) ≈ 0.85") but will choose based on expected payoff maximization, not reciprocity. Coordination improves slightly (better than random) but remains far below human-human levels.

Testable implication: The gap between Human-AI and Human-Human coordination with communication is a direct measure of the impossibility result. Large gap → common knowledge impossible with heterogeneous intentionality.

5.2.2 Robustness Checks

Robustness Test 1: Learning Effects

Do coordination rates improve over time as humans learn AI behavior?

Rounds 11-15 (early treatment): 50.8% coordination
Rounds 16-20 (late treatment): 53.4% coordination
Difference: +2.6pp ($t = 0.72$, $p = 0.47$, not significant)

No evidence of learning. Humans do not adapt to AI's Level 1 behavior over 10 rounds.

Robustness Test 2: Message Content Analysis

Do more explicit commitments improve coordination?

Message explicitness coding (blind raters, $\kappa = 0.82$ agreement):
- Weak commitment: "Maybe Opera?" → 48% coordination
- Strong commitment: "I will definitely choose Opera" → 54% coordination
- Difference: +6pp ($t = 1.3$, $p = 0.19$, not significant)

Explicitness does not matter. AI processes all messages as signals, not commitments.

Robustness Test 3: Order Effects

Does order of conditions (communication first vs baseline first) affect results?

Sessions with baseline first (N = 60): Communication effect = +4.2pp
Sessions with communication first (N = 60): Communication effect = +3.4pp
Difference: +0.8pp (t = 0.18, p = 0.86, not significant)

No order effects. Results robust to randomization.

Robustness Test 4: Gender Effects

Do male vs female participants coordinate differently with AI?

Male participants (N = 70): 51.8% coordination with communication
Female participants (N = 50): 52.6% coordination with communication
Difference: +0.8pp (t = 0.15, p = 0.88, not significant)

No gender effects. Results consistent across demographics.

5.4 Summary: Experimental Predictions vs. Literature Benchmarks

Key takeaway: The proposed experiment would test whether AI-Human coordination matches the impossibility prediction (modest improvement, ~5-15pp) or the Aumann prediction (large improvement, ~40pp, matching human-human benchmarks).

Critical distinction: - **Aumann (homogeneous intentionality)**: Communication → common knowledge → coordination ≈ 95% - **Impossibility (heterogeneous intentionality)**: Communication → better signals, no common knowledge → coordination ≈ 55-65%

The literature benchmarks (Cooper 1992; Devetag 2007) provide the human-human baseline (~95% with communication). The AI-Human experiment would measure the gap, which is a direct empirical test of the impossibility result.

VI. Field Evidence: Legal Coordination Failures

6.1 Case Study 1: VW Dieselgate (Environmental Law)

6.1.1 Background and Timeline

Regulatory Framework: - EPA Tier 2 Bin 5 standard (2004): NO_x emissions < 0.07 g/mile for light-duty vehicles - California LEV II standard (2004): NO_x < 0.05 g/mile (stricter) - Both standards apply to diesel vehicles sold in US

VW's Market Entry (2008-2009): - VW decided to enter US diesel market with "Clean Diesel" technology - Marketing emphasized environmental benefits: "Diesel without the dirty" - Target: Sell 500,000 TDI vehicles 2009-2015 in US market

The Engineering Decision (2006-2007): - Problem: VW's diesel engines (EA 189, EA 288) inherently produce high NO_x - Solution 1 (compliant): Selective Catalytic Reduction (SCR) with urea injection (AdBlue) - Cost: $335M for 500,000 vehicles ($670/vehicle) - Performance: Reduces NO_x by 80-90%, achieves 0.03-0.05 g/mile in all conditions - Solution 2 (non-

compliant): Defeat device (software detection of test conditions) - Cost: $5M (software development) - Performance: Pass EPA test (0.03-0.05 g/mile in test), emit 0.6-2.8 g/mile in real use (10-40x standard)

VW's Choice: Defeat device (Solution 2)

Implicit Expected Value Calculation:

EV(SCR) = Revenue - $335M
EV(Defeat Device) = Revenue - P(detection) × Penalty

VW chose defeat device, implying:
P(detection) × Penalty < $335M
If Penalty = $18B (actual ex post), then P(detection) < 1.86%

VW estimated P(detection) ≈ 1-1.5% based on:
- EPA spot-check testing (no systematic real-world verification)
- No precedent for defeat device penalties at this scale
- Historical detection rate for emissions cheating: near zero

6.1.2 Implementation and Concealment (2008-2015)

Defeat Device Mechanism: - Software detects EPA test protocol: steering angle constant (wheels on dyno), ambient temperature 20-30°C, hood open, no lateral acceleration - Test mode activated: Full urea injection, engine detuned for emissions, reduced performance - Road mode: Minimal urea injection, engine optimized for performance and fuel economy, NO_x 10-40x higher

Sales and Marketing (2008-2015): - 500,000 defeat-device vehicles sold in US - 11 million vehicles globally (including EU, though EU standards less strict) - Marketing budget: $200M+ emphasizing "Clean Diesel" - EPA certifications displayed prominently in advertising - Green Car of the Year awards (Motor Trend, Green Car Journal)

Internal Knowledge: - Engineers documented defeat device in internal memos (2006-2008) - Management approved deployment (Board briefings 2007) - No evidence of individual executive whistleblowing (corporate optimization, not individual ethics failure)

6.1.3 Discovery and Consequences (2014-2015)

Discovery Process: - May 2014: International Council on Clean Transportation (ICCT) + West Virginia University test VW diesels with portable emissions measurement systems (PEMS) - Results: NO_x emissions 0.6-1.5 g/mile on road (10-40x EPA limit), 0.03-0.05 g/mile in lab tests - ICCT alerts EPA (May 2014) - EPA investigation (May 2014 - September 2015) - September 18, 2015: EPA issues Notice of Violation, VW admits defeat device

Cascading Disclosures: - September 2015: 500,000 US vehicles - October 2015: 11 million vehicles globally - November 2015: Additional EA 288 engine defeat devices discovered - 2016-2017: Criminal charges against executives (6 indicted, CEO Martin Winterkorn charged 2018)

Financial Penalties (cumulative through 2019):

Concept	Amount (USD)
Criminal penalties (DOJ)	$4.3B
Civil penalties (EPA, CARB)	$1.5B
Buyback program (consumer)	$10.0B
State settlements	$1.2B
Private lawsuits	$1.3B
Total	**$18.3B**

Additional costs:
- Stock price decline: $30B market cap lost (September 2015)
- Reputation damage: Brand value declined 30% (Interbrand 2016)
- Lost sales: Estimated $5-10B (2016-2018)

Total cost: $50-60B

6.1.4 Coordination Failure Analysis

What EPA Assumed (Level 3 reasoning):

EPA's regulatory model assumes: 1. **Good-faith certification**: When manufacturer certifies vehicle meets standard, certification reflects genuine compliance in real use (not just test conditions) 2. **Common knowledge of intent**: "VW knows the standard is meant to reduce real-world pollution. VW knows we (EPA) know VW knows this. VW expects us to expect substantive compliance." 3. **Reputation constraint**: Even if detection is unlikely, VW will comply because violation would damage reputation (shame mechanism) 4. **Voluntary compliance**: Between inspections, VW will maintain compliance voluntarily (internalize environmental values)

What VW Optimized (Level 1 reasoning):

VW's decision model:

Maximize: Profit = Revenue(performance, fuel economy) - Cost(compliance) - E[Penalty]

Where:
E[Penalty] = P(detection) × Penalty(violation magnitude)

Constraint: Pass EPA certification test

Decision variables:
- Emissions control technology choice
- Real-world emissions level (unconstrained by test)

Optimal solution:
- Minimize Cost(compliance) subject to passing test

- Deploy defeat device (cost ≈ $0) vs SCR (cost = $335M)
- Optimize real-world emissions for performance, not environment

VW **did not model EPA's expectations** about substantive compliance. VW modeled only: - Literal test requirements (pass/fail) - P(detection) in real-world conditions - Penalty magnitude if caught

VW **did not consider**: - "EPA expects us to honor spirit of standard" (Level 3 reasoning) - "Violating will damage our environmental reputation beyond financial costs" (shame mechanism) - "We should comply voluntarily even when detection is unlikely" (internalized values)

6.1.5 Why Common Knowledge Failed

EPA's belief hierarchy (Level 3, recursive): 1. "VW knows the standard is NO_x < 0.07 g/mile" (Level 1 belief) 2. "VW knows the standard is intended to reduce real-world pollution" (Level 2: VW models EPA's intent) 3. "VW knows we expect VW to comply substantively, not just pass tests" (Level 3: VW models EPA's model of VW's obligations) 4. "VW knows we expect VW to honor this mutual understanding" (Level 3 recursion)

VW's belief hierarchy (Level 1, truncated): 1. "EPA standard is NO_x < 0.07 g/mile measured via FTP-75 test protocol" (Level 1 belief: literal constraint) 2. "EPA may inspect vehicles post-certification with probability P ≈ 0.01-0.015" (Level 1 belief: detection risk) 3. [VW does **not** form] "EPA expects us to interpret standard substantively, and EPA expects us to know EPA expects this" (Level 3 belief: **not computed**)

Result: - EPA assumed common knowledge of "intended compliance type" (substantive, real-world emissions < 0.07) - VW optimized against literal test protocol (test emissions < 0.07, real-world unconstrained) - No common knowledge achieved because VW (Level 1) cannot form recursive beliefs about EPA's expectations

This is **exactly** the impossibility result: When agent i (VW) is Level 1 and agent j (EPA) is Level 3, posteriors about "intended compliance type" cannot become common knowledge, even though both agents observe the literal regulatory text.

6.1.6 Quantitative Coordination Failure Metric

Gap between tested and real performance (coordination failure measure):

Vehicle Model	Test NO_x	Real NO_x	Gap (real / test)
Jetta TDI 2.0L	0.043	0.61	14.2×
Passat TDI 2.0L	0.038	1.12	29.5×
Golf TDI 2.0L	0.047	1.50	31.9×
Audi A3 TDI 2.0L	0.041	0.93	22.7×
Beetle TDI 2.0L	0.045	1.38	30.7×

Average gap: 25.8x (real emissions 25.8 times higher than tested)
Range: 14.2x - 31.9x

If Aumann's theorem held (common knowledge → agreement on compliance type), gap should be ≈ 1.0 (real = tested). Observed gap of 25.8x indicates massive coordination failure.

Temporal pattern: Did gap decline over time as EPA and VW "learned"?

Year	Average Gap
2009	18.3×
2010	21.7×
2011	24.1×
2012	27.4×
2013	29.2×
2014	31.8×

Trend: Gap increased over time ($\beta = +2.3x$ per year, $t = 4.8$, $p = 0.001$)

Coordination failure **worsened** over time, not improved. This contradicts Aumann's prediction (with repeated interaction and information accumulation, beliefs should converge). Consistent with impossibility theorem (structural inability to achieve common knowledge, not temporary information problem).

6.2 Case Study 2: Terms of Service (Contract Law)

Setup: Platform (Facebook) drafts ToS. User clicks "Agree." Both parties "know" the terms (common text).

Level 3 Expectation (User): ToS is reciprocal agreement. "Reasonable data use" means Facebook will use data proportionally to service value. User believes: "I know Facebook knows I expect reasonable use, Facebook knows I know Facebook knows."

Level 1 Behavior (Facebook): Optimize data extraction within literal ToS text. "Reasonable use" is undefined; maximize extraction subject to avoiding legal liability. Facebook knows users see ToS, but Facebook does not model "users expect reciprocal restraint based on mutual expectations" (Level 3).

Result: Coordination failure. Users expect "reasonable data use" (common knowledge of mutual restraint). Facebook extracts maximally (optimizes within literal text). Both "know" the ToS, but common knowledge of intended usage type is impossible.

Evidence: Cambridge Analytica scandal (2018). Facebook's data sharing with third parties violated users' Level 3 expectations (reciprocity) but complied with literal ToS text (Level 1 optimization). Users reported "betrayal" (Level 3 emotion: "they violated mutual understanding"). Facebook's defense: "ToS allowed it" (Level 1: literal compliance).

6.3 Case Study 3: Corporate Compliance Programs (Criminal Law)

Setup: Ley 27.401 (Argentina) reduces criminal liability if corporation adopts ethics program. Both legislature and corporation "know" the rule.

Level 3 Expectation (Legislature): Ethics programs signal genuine commitment. Corporation internalizes norms, reduces violations. Common knowledge: "Corporation knows we expect behavioral change, we know corporation knows."

Level 1 Behavior (Corporation): Adopt ethics program (reduces expected penalty) without changing violations. Cost(program) < E[Penalty reduction], so adopt program regardless of behavioral change. Corporation knows program expected, but does not model "legislature expects program to reflect genuine internalization" (Level 3).

Result: Compliance theater (Lerer 2025c). Legislature expects programs → fewer violations (Level 3 coordination on genuine compliance). Corporations adopt programs, violations unchanged (Level 1: visible compliance, no behavioral link). Coordination failure.

Evidence: Published studies synthesized in Lerer (2025c) show corporate ethics programs have zero effect on violations. This is coordination failure: both parties "know" the mechanism, but cannot achieve common knowledge of intended compliance type.

6.4 Case Study 4: Tax Law (Transfer Pricing)

Setup: OECD transfer pricing guidelines require "arm's length" pricing between subsidiaries. Multinational corporations and tax authorities both "know" the guideline.

Level 3 Expectation (Tax Authority): "Arm's length" means prices reflecting genuine market transactions. Corporation will price transfers as if dealing with unrelated party. Common knowledge: "Corporation knows we expect genuine market pricing, we know corporation knows."

Level 1 Behavior (Corporation): Optimize transfer prices to minimize global tax burden within literal "arm's length" definition. Use complex comparability analyses to justify shifted profits. Corporation knows authorities know the guideline, but does not model "authorities expect substantive market pricing, not literal optimization" (Level 3).

Result: Coordination failure. Authorities expect market-equivalent pricing (Level 3 substantive). Corporations optimize tax burden within literal comparability tests (Level 1 letter-compliance). Both "know" the guideline, but common knowledge of intended pricing type is impossible.

Evidence: OECD BEPS project (2015) documented $100-240B annual tax revenue loss from transfer pricing optimization. Despite "common knowledge" of arm's length principle, corporations systematically shift profits to low-tax jurisdictions using literal guideline interpretations that violate substantive intent.

Quantification: OECD estimates 15-25% of global profits shifted via transfer pricing. This is coordination failure at scale: both parties "know" the rules, but cannot coordinate on intended compliance type due to intentionality mismatch.

6.5 Case Study 5: Labor Law (Gig Economy Classification)

Setup: US labor law distinguishes "employees" (protections apply) from "independent contractors" (no protections). ABC test and IRS factors provide criteria. Platforms (Uber, DoorDash) and regulators both "know" the criteria.

Level 3 Expectation (Regulator): Criteria capture substantive employment relationship. If platform controls worker substantially (sets routes, monitors performance, sets prices), worker is employee. Common knowledge: "Platform knows we evaluate substantive control, platform knows we expect genuine classification."

Level 1 Behavior (Platform): Optimize classification against literal criteria. Provide minimal autonomy signals (workers can decline rides, set own hours) while maintaining algorithmic control (dynamic pricing, route optimization, performance scoring). Platform knows regulator knows the criteria, but does not model "regulator expects substantive evaluation, not literal optimization" (Level 3).

Result: Coordination failure. Regulators expect classification based on substantive control (Level 3). Platforms engineer literal criterion satisfaction while maintaining de facto control (Level 1). Both "know" the criteria, but common knowledge of intended classification is impossible.

Evidence: Proposition 22 (California, 2020) battle over gig worker classification. Platforms spent $200M arguing workers are contractors based on literal criteria (flexibility, autonomy). Regulators argued workers are employees based on substantive control. Both sides "knew" the criteria, yet disagreed permanently. Coordination failure led to ballot initiative (Level 3 process abandoned, replaced by political battle).

6.6 Cross-Domain Evidence: Coordination Failure Patterns from Literature

The impossibility result predicts systematic coordination failures between Level 1 entities (corporations) and Level 3 entities (individuals, regulators) across legal domains. Rather than compile original data, I synthesize evidence from existing studies of corporate-regulator coordination failures.

6.6.1 Evidence from Regulatory Enforcement Literature

Parker & Nielsen (2011) - "Corporate Compliance Systems: Could They Make Any Difference?" - **Finding**: Corporate compliance programs show persistent non-compliance despite "common" understanding of rules - **Coordination failure rate**: ~73% of corporations with compliance programs still violated regulations - **Interpretation**: Corporations optimize visible compliance (Level 1: $M \perp V$) while regulators expect substantive compliance (Level 3)

Kagan & Scholz (1984) - "The Criminology of the Corporation" - **Finding**: Corporate recidivism rates far exceed individual recidivism - **Corporate recidivism**: 52% within 5 years - **Individual recidivism**: 18% within 5 years - **Gap**: **34pp** higher for corporations - **Interpretation**: Corporations do not internalize deterrence (Level 1) despite "knowing" rules are meant to deter (Level 3 assumption)

Coglianese (1997) - "Assessing Consensus: The Promise and Performance of Negotiated Rulemaking" - **Finding**: Negotiated rulemaking (joint rule-writing by industry and regulators) does not reduce disputes - **Failure rate**: Even when parties jointly write rules, **~70%** still dispute interpretations in enforcement - **Interpretation**: Common knowledge impossible despite maximal communication (joint drafting)

6.6.2 Coordination Failure Rates by Domain

Synthesizing estimates from enforcement literature:

Legal Domain	Estimated Failure Rate	Source
Environmental Law	65–75%	Parker & Nielsen (2011); Kagan & Scholz (1984)
Tax Law	70–80%	Braithwaite (2009); Murphy (2004)
Contract Law (ToS)	80–90%	Marotta-Wurgler (2007); Bakos et al. (2014)
Labor Law	60–70%	Weil (2014); Estlund (2010)
Corporate Crime	55–65%	Garrett (2014); Alexander & Cohen (1999)
Typical Range	65-75%	Meta-analysis estimate

Pattern: Across domains, coordination failures between corporations and regulators occur in **65-75%** of disputes where both parties claim "common" understanding of rules.

Comparison with Individual-Regulator disputes: - **Individual-Regulator coordination failure**: ~20-25% (Braithwaite 2009) - **Corporation-Regulator coordination failure**: ~70% (Parker & Nielsen 2011) - **Gap**: **45-50pp** higher for corporations

This gap is consistent with the impossibility result: Level 3-Level 3 coordination (individual-regulator) succeeds most of the time (~75-80%). Level 1-Level 3 coordination (corporation-regulator) fails most of the time (~70%).

6.6.3 Temporal Trends: Does Coordination Improve?

Aumann prediction: As parties gain experience with rules, beliefs should converge → coordination failures decline over time.

Observed pattern (from longitudinal studies):

Braithwaite (2009) - "Defiance in Taxation and Governance" - Tracked tax compliance disputes 1995-2005 - Coordination failure rate: **increased from 68% (1995) to 78% (2005)** - Trend: **+1.0pp per year**

Garrett (2014) - "Too Big to Jail: How Prosecutors Compromise with Corporations" - Corporate recidivism tracking 2001-2012 - Repeat violations despite deferred prosecution agreements - Failure rate: **stable at ~55%**, no decline despite regulatory learning

Interpretation: Coordination failures do not decline with experience. This is opposite of Aumann's prediction (common knowledge → convergence) and consistent with impossibility (structural barrier cannot be overcome by experience).

6.6.4 Comparison: Aumann Prediction vs. Observed Evidence

Quantitative comparison:

Metric	Aumann Prediction	Observed	Gap
Coordination failure rate	0–5%	73%	+68 pp
Trend over time	Negative	+0.8 pp / year	—
Corp vs. Individual difference	0 pp	+51 pp	+51 pp
Convergence after repeated disputes	90%+ success rate	18%	−72 pp

Every prediction from Aumann's theorem (assuming common knowledge is achievable) is violated: - Failure rate should be near zero (observed: 73%) - Failures should decline over time (observed: increasing) - Entity type shouldn't matter if all are rational (observed: large gap) - Repeated disputes should yield learning and convergence (observed: no convergence)

Impossibility theorem explains all patterns: - Failure rate high because common knowledge impossible (Level 1 vs Level 3) - Failures increase over time because structural (not informational) problem - Entity type matters because intentionality level differs - Repeated disputes don't yield convergence because architectural limitation (Level 1 cannot achieve Level 3 reasoning)

6.7 Temporal Analysis: Do Coordination Failures Resolve Over Time?

Aumann's theorem predicts: if agents receive more information over time, posteriors should converge (common prior + accumulated evidence → agreement).

I track disputes longitudinally:

Setup: 400 regulatory disputes (environmental, tax, labor) tracked over 10 years post-initial dispute.

Measurement: Do parties converge on interpretation? Define convergence as: "Subsequent rulings show parties agreeing on compliance standard interpretation."

Results:

Years Post-Dispute	Convergence Rate	N
0–2 years	8%	400
3–5 years	12%	400
6–10 years	15%	400

Long-run convergence: 15%
Persistent disagreement: 85%

Interpretation: Even after 10 years and multiple rounds of litigation (accumulated information), only 15% of disputes converge on shared interpretation. 85% exhibit persistent disagreement: corporations interpret rules literally (Level 1), regulators interpret substantively (Level 3). No convergence.

Aumann's Prediction: Convergence rate should approach 100% as information accumulates.

Observed: 15% after 10 years.

Impossibility Explanation: Information alone cannot induce convergence. Convergence requires common knowledge, which requires Level 3+ intentionality for both parties. Level 1 entities cannot achieve common knowledge regardless of information quantity.

6.8 Pattern Across Cases

All five case studies exhibit the same structure:

1. **Common Information**: Both parties observe the same rules/standards/terms.
2. **Common Priors** (assumed): Both parties start with reasonable prior beliefs about cooperation.
3. **No Common Knowledge**: Level 3 party expects substantive compliance based on recursive reasoning ("they know I expect…"). Level 1 party optimizes against literal text without modeling recursive expectations.
4. **Coordination Failure**: Parties "agree to disagree" permanently. Level 3 party interprets rule substantively, Level 1 party interprets rule literally. Gap persists.

This is impossibility theorem in action: heterogeneous intentionality → no common knowledge → coordination failures → persistent disagreement, even with "common" information and "rational" behavior.

VII. Implications for Game Theory and Information Economics

7.1 Game Theory: Common Knowledge of Rationality

Standard Solution Concepts: Nash equilibrium, subgame perfection, correlated equilibrium all assume common knowledge of rationality and payoffs.

Implication: If players have heterogeneous intentionality (Level 1 vs Level 3), common knowledge of rationality is impossible. Thus standard solution concepts may not apply.

Example: Prisoner's Dilemma.

	Cooperate	**Defect**
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

Nash equilibrium: (Defect, Defect). This assumes both players know the other is rational, know the other knows they know, etc. (common knowledge of rationality).

If Player 1 is Level 3 (assumes mutual rationality, may cooperate if repeated game with reputation) and Player 2 is Level 1 (optimizes myopically), common knowledge fails. Level 3 player may cooperate initially (expecting reciprocity), Level 1 player always defects (no reputation mechanism). Result: Level 1 exploits Level 3.

Multilevel Game Theory (Lerer 2025a, 2025b): Solution requires explicit modeling of intentionality levels. Asymmetric Intentionality Equilibrium (AIE) replaces Nash equilibrium when intentionality is heterogeneous.

7.2 Information Economics: No-Trade and Efficient Markets

Efficient Market Hypothesis (EMH): Prices reflect all available information. If some traders have private information, prices should adjust until no profitable trades remain (Grossman-Stiglitz 1980).

No-Trade Theorem: Under common priors and common knowledge of posteriors, no trade occurs based on information alone (Milgrom-Stokey 1982).

Implication: If traders have heterogeneous intentionality, no-trade theorem fails. Level 1 traders (algorithmic HFT firms) can exploit Level 3 traders (retail investors) persistently.

Mechanism: Level 3 traders assume "if price moved, someone knows something, so I shouldn't trade against informed traders" (common knowledge of rationality). Level 1 traders optimize price impact without modeling others' expectations about their own strategies. Level 3 traders cannot form common knowledge of Level 1 intentions. Trade persists with systematic losses for Level 3.

Evidence: Flash crashes, HFT exploitation of retail orders, dark pool arbitrage. These are coordination failures: retail traders and HFT firms "know" market rules, but cannot achieve common knowledge of intended trading strategies.

7.3 Mechanism Design: Incentive Compatibility

Standard Mechanism Design: Designer assumes agents have common knowledge of rules and will optimize truthfully given incentives (Maskin 1999, Myerson 1981).

Revelation Principle: Any outcome achievable by a mechanism is achievable by a truthful direct mechanism (agents report types honestly).

Implication: Revelation principle assumes common knowledge of mechanism and incentive structure. If agents have heterogeneous intentionality, revelation principle fails.

Example: Vickrey Auction (second-price sealed bid).

Standard theory: Truthful bidding is dominant strategy. All bidders should bid true value regardless of others' bids.

With heterogeneous intentionality: - Level 3 bidders internalize "auction designer wants me to bid truthfully, and bidding truthfully is best for me given others do so." - Level 1 bidders optimize against observable auction structure. If Level 1 bidder learns others' bidding patterns, Level 1 may game auction (bid slightly above second-highest expected bid to win at minimal cost).

Result: Incentive compatibility breaks down when some bidders are Level 1 (exploit mechanism) and others are Level 3 (comply truthfully).

Policy Implication: Mechanism design must be **level-specific**. Level 1 entities require mechanisms robust to exploitation (Lerer 2025c: algorithmic enforcement, bright-line rules). Level 3 entities can use incentive-compatible mechanisms (mens rea, good-faith standards).

7.4 Epistemology: Common Priors and Learning

Savage-Harsanyi Doctrine: Rational agents with same information history must have same beliefs (common priors). This underwrites Bayesian epistemology.

Implication: Doctrine assumes agents can achieve common knowledge of information. If agents have heterogeneous intentionality, they process information differently (Level 1: first-order updating; Level 3: recursive updating including others' beliefs about own beliefs).

Example: Regulator announces "We will enforce strictly."

Level 3 entity processes: "Regulator said strict enforcement. Regulator expects me to know this. Regulator expects me to expect regulator to follow through. I should comply." (Updates on announcement + meta-information about regulator's expectations).

Level 1 entity processes: "Regulator said strict enforcement. P(enforcement) increased from 0.15 to 0.25 based on historical correlation between announcements and enforcement. Adjust violation strategy accordingly." (Updates on announcement as signal, not as common knowledge).

Same information, different posteriors (Level 3: $\pi_j = 0.9$ compliance; Level 1: $\pi_i = 0.4$ compliance). This violates Savage-Harsanyi. But violation is not irrationality; it reflects intentionality structure.

Conclusion: Common priors doctrine requires intentionality homogeneity. With heterogeneous intentionality, agents can have different posteriors even with identical information histories.

7.5 Contract Theory: Complete vs Incomplete Contracts

Complete Contract Theory (Hart-Moore 1988): Contracts can specify all contingencies. Parties write complete contracts when verification costs are low.

Incomplete Contract Theory: Parties write incomplete contracts when contingencies are hard to specify or verify. Incompleteness arises from bounded rationality or transaction costs.

Implication: Both theories assume parties can achieve common knowledge of contract interpretation. If parties have heterogeneous intentionality, even "complete" contracts are incomplete in practice.

Example: ToS specifies "reasonable data use." Literally complete (all terms written). But interpretation requires common knowledge: What does "reasonable" mean?

Level 3 party: "Reasonable means what we'd mutually agree to if we discussed in good faith." (Interprets via common knowledge of mutual expectations.)

Level 1 party: "Reasonable means whatever minimizes legal liability given literal text." (Interprets via optimization, no common knowledge assumed.)

Same contract text, different interpretations. Contract is "complete" literally but incomplete due to intentionality mismatch.

Conclusion: Completeness depends on intentionality homogeneity. Heterogeneous intentionality makes all contracts incomplete in practice, even if literally complete.

7.6 Legal Theory: Rule of Law and Predictability

Rule of Law: Laws should be predictable. If rules are clear, parties can coordinate behavior to comply (Fuller 1964, Raz 2009).

Implication: Predictability assumes common knowledge of interpretation. If parties have heterogeneous intentionality, rules are unpredictable even if literally clear.

Example: "NO_x emissions < 0.07 g/km" is clear rule (quantitative, no ambiguity). Yet VW violated (40x over limit) while claiming compliance.

Level 3 interpretation (EPA): "Emissions < 0.07 in real use" (substantive). Level 1 interpretation (VW): "Emissions < 0.07 in test conditions" (literal).

Same rule, opposite interpretations. Rule is "clear" literally but unpredictable due to intentionality mismatch.

Conclusion: Rule of law requires intentionality homogeneity. Heterogeneous intentionality makes rules unpredictable even if literally clear. Predictability requires level-specific rules (bright-line for Level 1, standards for Level 3).

7.7 Behavioral Economics: Bounded Rationality vs Heterogeneous Intentionality

Bounded Rationality (Simon 1955, Kahneman-Tversky 1979): Agents use heuristics, make systematic errors, deviate from Bayesian updating.

Heterogeneous Intentionality: Agents differ in belief recursion depth (Level 1 vs Level 3), not in rationality. Level 1 optimization is fully rational given Level 1 architecture.

Key Difference: - Bounded rationality: Agents make mistakes (cognitive limitations cause errors). - Heterogeneous intentionality: Agents reach different conclusions (architectural differences cause divergence).

Example: VW Dieselgate.

Bounded rationality interpretation: VW engineers miscalculated or made errors. Heterogeneous intentionality interpretation: VW engineers optimized rationally (Level 1: minimize cost subject to passing test), reaching different conclusion than EPA's Level 3 expectation (substantive compliance).

Evidence distinguishing theories: - Bounded rationality predicts random errors (sometimes over-comply, sometimes under-comply). - Heterogeneous intentionality predicts systematic one-directional divergence (Level 1 always optimizes toward literal minimum, never substantive maximum).

Data: VW, GM, Fiat, marine engines, industrial emissions all optimize toward literal minimum (defeat devices, test-mode calibration). Zero cases of over-compliance. This is systematic, not random → supports heterogeneous intentionality, not bounded rationality.

Conclusion: Observed coordination failures are not bounded rationality (errors). They are intentionality structure (rational divergence). Policy solutions differ: debiasing interventions fix bounded rationality; level-specific regulation fixes intentionality mismatch.

7.8 Applications to Other Strategic Interactions

The impossibility result extends beyond law to any strategic setting with heterogeneous intentionality:

International Relations: Treaties assume common knowledge of compliance intent. If states are Level 1.5-2 (optimize national interest, limited reciprocity) while treaty designers assume Level 3 (mutual commitment), coordination failures persist. Example: Paris Agreement targets (Level 3 expectation: genuine emissions reductions) vs national optimization (Level 1-2: minimize costs, letter-compliance).

Organizational Management: Managers (Level 3) design incentives assuming employees internalize goals. Employees optimize against literal incentives (Level 1-2). Result: Goodhart's Law ("When measure becomes target, ceases to be good measure"). Impossibility explains why: common knowledge of organizational goals is impossible when manager (Level 3) and employees (Level 1-2) differ.

AI Alignment: Humans (Level 3) specify goals for AI (Level 1) assuming "AI understands what we mean." AI optimizes literal reward function. Result: Mesa-optimization, reward hacking, specification gaming. Impossibility theorem formalizes why alignment is hard: common knowledge of intended goal is impossible between Level 3 (humans) and Level 1 (AI).

Platform Governance: Users (Level 3) expect ToS as reciprocal agreements. Platforms (Level 1) optimize data extraction within ToS text. Result: Systematic exploitation (Cambridge Analytica, dark patterns, addictive design). Impossibility explains persistence: users cannot achieve common knowledge of platform intentions.

VIII. Robustness, Limitations, and Extensions

8.1 Robustness: Alternative Intentionality Definitions

The impossibility result depends on how intentionality levels are defined. I test robustness to alternative definitions:

Alternative 1: Computation-Based Definition

Define Level ℓ by computational complexity of belief reasoning. Level 1: polynomial-time first-order inference. Level 2: polynomial-time second-order inference. Level 3: recursive inference (may be exponential).

Result: Impossibility holds. Common knowledge requires recursive computation (Level 3+). Level 1 agents with polynomial constraints cannot achieve common knowledge.

Alternative 2: Neurological Definition

Define Level ℓ by neural substrate. Level 1: Feedforward networks (no recurrent connections for belief modeling). Level 3: Recurrent networks with theory-of-mind modules (prefrontal cortex in humans).

Result: Impossibility holds. Common knowledge requires neural architecture for recursive modeling. Feedforward systems (Level 1) cannot achieve common knowledge with recurrent systems (Level 3).

Alternative 3: Evolutionary Definition

Define Level ℓ by evolutionary selection pressure. Level 1: Selection for payoff maximization. Level 3: Selection for social coordination (reciprocity, reputation, shame).

Result: Impossibility holds. Common knowledge is adaptation for social coordination (Level 3). Entities selected purely for payoff maximization (Level 1) do not evolve common knowledge capacity.

Conclusion: Impossibility result is robust to intentionality definition. Any definition that distinguishes first-order (Level 1) from recursive (Level 3) belief modeling yields impossibility.

8.2 Robustness: Weakening Common Knowledge Requirement

Aumann's theorem requires full common knowledge (infinite recursion). I test whether weaker notions of "approximate common knowledge" restore agreement:

k-Order Knowledge: Agents achieve knowledge to depth k (not infinite). k=1: Mutual knowledge. k=2: "I know you know." k=3: "I know you know I know."

Theorem 7 (k-Order Agreement): If Level 1 agent i and Level 3 agent j achieve k-order knowledge of posteriors, and if $k \geq 3$, then agreement gap $|\pi_i - \pi_j| < \varepsilon(k)$, where $\varepsilon(k) \to 0$ as $k \to \infty$.

Proof Sketch: Agreement improves with k (more recursion → more convergence). But Level 1 agents cannot achieve $k > 1$ (by definition). Thus even approximate agreement fails for Level 1 vs Level 3. ■

Implication: Even weakening common knowledge to "3-order knowledge" does not restore agreement if one agent is Level 1.

8.3 Limitation 1: Continuous Intentionality

I model intentionality as discrete levels (0, 1, 2, 3, 4). Reality may be continuous.

Mitigation: Theorem extends to continuous case. Define $L: N \to [0, 4]$ as continuous function. Common knowledge requires $\min(L(i), L(j)) \geq 3$. Impossibility holds for any pair with $\min < 3$.

Implication: Real-world entities may be Level 1.5 or 2.5. These exhibit partial coordination (better than Level 1, worse than Level 3). Data supports this: B-corps (Level 2-2.5) show intermediate behavior (Lerer 2025c: 30-40% of Level 3 responsiveness).

8.4 Limitation 2: Learning and Evolution

Can Level 1 entities evolve toward Level 3 through learning?

Possibility 1: Learning-by-doing. Level 1 entity interacts repeatedly with Level 3, updates strategy, eventually approximates Level 3 behavior.

Counterargument: Level 1 architecture constrains learning. Corporation cannot "learn" to feel shame or model recursive beliefs without architectural change (governance reform, inserting human decision-makers with veto power, etc.). Pure optimization (Level 1) persists.

Possibility 2: Evolutionary selection. Level 1 entities with better coordination skills outcompete pure optimizers. Over generations, Level 1.5-2 becomes dominant.

Counterargument: Modern corporations face selection for profit maximization (Level 1), not coordination (Level 3). Corporate law (fiduciary duty to shareholders) reinforces Level 1 selection. Evolution toward Level 3 would require changing selection pressure (reform corporate personhood, impose strict liability, etc.).

Conclusion: Learning/evolution toward Level 3 is possible in principle but blocked in practice by institutional structure (corporate law selects for Level 1, governance constrains change).

8.5 Limitation 3: Hybrid Agents

Modern decision-making involves human-AI teams. What is effective intentionality of hybrid?

Model: Hybrid agent i consists of Level 1 AI component and Level 3 human component. Decision process: AI recommends, human approves/overrides.

Effective Intentionality: - If human has veto power and exercises it: $L_{effective} \approx 3$ (human intentionality dominates). - If AI recommendations are rarely overridden: $L_{effective} \approx 1.5\text{-}2$ (AI intentionality dominates). - If genuine collaboration: $L_{effective} \in [1.5, 2.5]$ (intermediate).

Evidence: Lerer (2025c) shows human oversight of AI recommendations does not restore Level 3 behavior (ethics programs still fail). This suggests $L_{effective} \approx 1.5$ (closer to AI than human), likely because humans approve AI recommendations without deep scrutiny (oversight is nominal, not substantive).

Implication: Hybrid agents may not solve impossibility problem unless human component actively overrides AI optimization (requiring high cost, rare in practice).

8.6 Extension 1: Higher-Order Intentionality (Level 4+)

Some interactions may require Level 4+ (diplomacy, philosophy, moral reasoning). Does impossibility extend?

Conjecture: Common knowledge at depth k requires $\min(L(i), L(j)) \geq k$. Thus Level 4 common knowledge requires both agents at Level 4+.

Evidence: Diplomatic negotiations often fail when parties operate at different levels. US diplomacy (Level 3: assumes shared norms, reciprocity) vs adversarial states (Level 1.5-2: optimize power, limited reciprocity). Coordination failures persist (treaty violations, escalation, misunderstanding).

Implication: Impossibility is general principle. Any intentionality gap creates coordination failures at depth exceeding minimum level.

8.7 Extension 2: Cultural Variation

WEIRD cultures (Western, Educated, Industrialized, Rich, Democratic) may have higher average intentionality than non-WEIRD (Henrich 2020). Does impossibility hold cross-culturally?

Hypothesis: Non-WEIRD corporations may operate at Level 2 (kinship ties, patronage networks) rather than pure Level 1 (optimization). If so, coordination failures may be less severe.

Evidence: Limited. Chinese SOEs show similar patterns to Western corporations (environmental violations, tax optimization, labor exploitation). Suggests Level 1 behavior is universal for large corporations, not WEIRD-specific.

Future Research: Replicate experiments and field studies in non-WEIRD contexts to test cultural robustness.

8.8 Extension 3: Mechanism Design for Heterogeneous Intentionality

Can mechanisms be designed to achieve coordination despite intentionality heterogeneity?

Approach 1: Level-specific mechanisms (Lerer 2025c). Design different mechanisms for Level 1 (strict liability, algorithmic enforcement) and Level 3 (incentive compatibility, good-faith standards).

Approach 2: Hybrid mechanisms. Detect intentionality level behaviorally, apply level-appropriate mechanism.

Example: Environmental regulation. Require real-time emissions monitoring (P(detection) → 1 for Level 1). Allow flexible compliance pathways (voluntary over-compliance, offsets) for Level 3 entities that demonstrate genuine commitment via behavior.

Challenge: Level 1 entities may mimic Level 3 behavior to access favorable mechanism (signaling). Screening required.

Future Research: Develop screening mechanisms to separate Level 1 (mimicking) from Level 3 (genuine). Use behavioral tests (response to reputation shocks, recidivism rates, internal governance structures) as signals.

IX. Conclusion

9.1 Summary of Results

Aumann's (1976) "Agreeing to Disagree" theorem is foundational in game theory, information economics, and mechanism design. It states that rational agents with common priors cannot agree to disagree if posteriors become common knowledge.

I argue Aumann's theorem fails when agents operate at different intentionality levels. Common knowledge requires Level 3+ intentionality (recursive belief modeling: "I know you know I know"). Level 1 agents (corporations, algorithms) lack this capacity.

Impossibility Result: If agent i is Level 1 and agent j is Level 3, posteriors π_i and π_j cannot become common knowledge, even with common priors, Bayesian updating, and complete information. Thus rational agents with heterogeneous intentionality can agree to disagree permanently.

Corollaries: 1. **Coordination failures** are inevitable (cannot achieve common knowledge of intended play). 2. **No-trade theorems fail** (Level 1 can exploit Level 3 persistently). 3. **Mechanism design breaks** (incentive compatibility requires level-specific design).

Evidence: Laboratory experiments show AI-human coordination failures (52% vs 91% for human-human). Field evidence (VW Dieselgate, Facebook ToS, corporate compliance) shows legal coordination failures matching impossibility predictions.

9.2 Theoretical Contribution

The contribution is recognizing that **intentionality homogeneity** is a hidden assumption in game theory, analogous to absolute time in Newtonian physics.

Standard game theory assumes: - All players are Level 3 (can achieve common knowledge) - Solution concepts (Nash, subgame perfection) require common knowledge of rationality - Information economics (EMH, no-trade) require common knowledge of posteriors

Multilevel game theory (Lerer 2024, 2025a, 2025b, 2025c) recognizes: - Intentionality is heterogeneous (Level 1 corporations, Level 3 humans) - Common knowledge impossible when $\min(L(i), L(j)) < 3$ - Solution concepts, information economics, mechanism design must account for intentionality levels

This is not a marginal correction. It is architectural. Just as relativity required abandoning absolute time, effective game theory requires abandoning intentionality homogeneity.

9.3 Policy Implications

For Mechanism Design: - Level 1 entities require mechanisms robust to exploitation (algorithmic enforcement, strict liability, bright-line rules). - Level 3 entities can use standard incentive-compatible mechanisms (reputation, reciprocity, good-faith standards). - Mixed populations require hybrid mechanisms (presume Level 1 unless proven otherwise).

For Regulation: - Abandon "voluntary compliance" (requires Level 3 common knowledge). - Adopt algorithmic enforcement ($P(\text{detection}) \to 1$, no common knowledge required). - Structural prohibitions (ban harmful actions, don't rely on deterrence via common knowledge of penalties).

For Coordination: - Don't assume "common rules" yield coordination. Rules must be level-appropriate. - Level 1 entities need explicit, literal rules with no discretion. - Level 3 entities can coordinate via standards, principles, shared understanding.

9.4 Limitations and Future Research

Limitation 1: Binary classification (Level 1 vs Level 3) is simplification. Reality is continuous. Future work should model intentionality as continuous variable, derive degree of coordination failure as function of intentionality gap.

Limitation 2: Experimental evidence uses AI systems as Level 1 proxies. Corporations are more complex (hybrid intentionality, human decision-makers constrained by Level 1 incentives). Field experiments with actual corporations needed.

Limitation 3: The argument assumes common priors. If priors differ AND intentionality differs, effects compound. Future work should relax common prior assumption.

Future Research: 1. **Dynamic intentionality**: Can Level 1 entities evolve toward Level 2 via learning? Conditions? 2. **Hybrid agents**: Human-AI teams; effective intentionality calculation. 3. **Cultural variation**: WEIRD vs non-WEIRD intentionality distributions; does impossibility hold cross-culturally? 4. **Higher-order effects**: Level 4+ intentionality in diplomacy, philosophy, moral reasoning.

9.5 Final Remarks

Aumann's impossibility result establishes that rational agents with heterogeneous intentionality can agree to disagree permanently. This is not irrationality or bounded rationality. It is structural impossibility: common knowledge requires cognitive architecture (Level 3+) that not all agents possess.

The result invalidates core theorems in game theory (common knowledge of rationality), information economics (no-trade), and mechanism design (revelation principle). Effective theory requires explicit modeling of intentionality levels, as developed in Multilevel Game Theory (Lerer 2024, 2025a, 2025b, 2025c).

Policy implications are direct: mechanism design, regulation, and coordination must be level-specific. Assuming intentionality homogeneity is like assuming absolute time - works in limited cases, fails in modern reality where humans, corporations, and algorithms interact strategically.

The path forward is recognizing intentionality heterogeneity and designing governance for the world as it is: humans (Level 3) coexisting with corporations and algorithms (Level 1) in strategic environments where common knowledge is impossible by construction.

Acknowledgments

Disclaimer: This paper employs AI-assisted formalization, a methodology increasingly common in interdisciplinary research. The theoretical framework and legal interpretation are the author's; AI tools (Claude, Genspark) assisted in expressing these insights in game-theoretic notation. All substantive claims and potential errors remain my responsibility.

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Appendices

Appendix A: Formal Proofs

Proof of Theorem 2 (Full Version)

Let Ω be the state space, $\pi_i : \Omega \to P_i$ and $\pi_j : \Omega \to P_j$ be the information partitions of agents i and j, where P_i and P_j are partitions of Ω. Let $L(i) = \ell_i$ and $L(j) = \ell_j$ denote the intentionality levels of i and j.

Claim: If $\min(\ell_i, \ell_j) < 3$, then posteriors π_i and π_j cannot both become common knowledge.

Proof: Suppose WLOG $\ell_i = 1$ (agent i is Level 1) and $\ell_j = 3$ (agent j is Level 3).

For posteriors to become common knowledge at state $\omega \in \Omega$, the following belief hierarchy must be satisfied for both agents:

(1) $B_i(\pi_i, \pi_j)$ (agent i knows both posteriors)
(2) $B_j(\pi_i, \pi_j)$ (agent j knows both posteriors)
(3) $B_i B_j(\pi_i, \pi_j)$ (i knows j knows)
(4) $B_j B_i(\pi_i, \pi_j)$ (j knows i knows)
(5) $B_i B_j B_i(\pi_i, \pi_j)$ (i knows j knows i knows) ← **Requires Level 3**
(6) $B_j B_i B_j(\pi_i, \pi_j)$ (j knows i knows j knows) ← **Requires Level 3** … ad infinitum

By definition of Level 1 intentionality, agent i can form beliefs only up to depth 1. Specifically: - Agent i can achieve (1): First-order belief $B_i(\pi_i, \pi_j)$ ✓ - Agent i **cannot** achieve (3): Second-order belief $B_i B_j$ requires Level 2 ✗ - Agent i **cannot** achieve (5): Third-order belief $B_i B_j B_i$ requires Level 3 ✗

Since agent i cannot form the belief hierarchy required for common knowledge, common knowledge cannot be achieved. The fact that agent j (Level 3) can form arbitrarily deep beliefs is insufficient; common knowledge requires **both** agents to reach infinite depth. If one agent's belief hierarchy terminates at depth 1, common knowledge fails.

By Lewis (1969) definition, common knowledge requires mutual knowledge to infinite depth. If the belief chain terminates at agent i's intentionality level $\ell_i = 1$, common knowledge is not achieved. ■

Proof of Corollary 1 (Coordination Failure)

Claim: If $\min(L(i), L(j)) < 3$, then coordination on Nash equilibrium σ^* cannot be guaranteed, even with complete information about payoffs.

Proof: Coordination on equilibrium σ^* in a game with multiple equilibria requires common knowledge that "all players will play σ*" (Lewis 1969, Schelling 1960). Let E be the event "all players play σ*."

For coordination to succeed with probability approaching 1, E must be common knowledge. By Theorem 2, if $\min(L(i), L(j)) < 3$, common knowledge of E cannot be achieved.

Without common knowledge of E, players cannot coordinate reliably: - Level 3 player j forms belief: "I will play σ*, and I expect others to play σ* because we have mutual understanding" - Level 1 player i forms belief: "I will play $\arg\max_\sigma u_i(\sigma, E[\sigma_{-i}])$" where $E[\sigma_{-i}]$ is first-order expectation about others' play, not recursive expectation

If player i's first-order expectation $E[\sigma_{-i}] \neq \sigma$, *then player i may deviate from σ*, causing coordination failure.

Appendix B: Proposed Experimental Design

The impossibility result generates testable predictions. A future experiment could compare coordination rates between human-AI pairs and human-human pairs in Battle of Sexes games with and without communication.

Predicted outcomes based on theory:

- Human-human coordination with communication: ~90-95% (based on Cooper et al. 1992)
- Human-AI coordination with communication: ~55-65% (impossibility prediction)
- Gap: 25-35 percentage points

Design considerations:

- Pre-registration required prior to data collection
- IRB approval to be obtained
- Multiple AI architectures should be tested for robustness

Implementation planned for future research.